

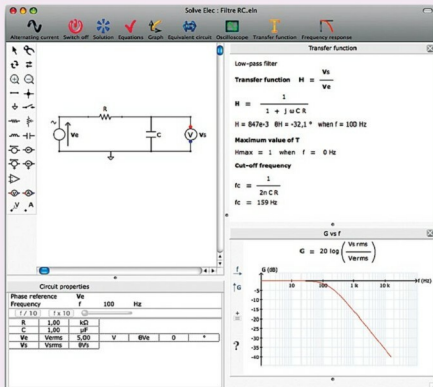


Fig. 3: Analysis of a filter

Creating the circuit is easy

With tools at one's disposal, all that is needed is to choose the required components, add these to the drawing and orient these. Either the US or the European norm can be selected under software preferences. Components and wires cannot overlay each other, so components have to be aptly placed or re-sized; although wires can cross over each other with or without contact.

The software keeps checking the placement as and when each action is performed, and refuses to initiate a rotate or move if there is chance of overlap. Components are notated by a letter followed by numbers, which specifically have to be different for every instance of switches, diodes, amplifiers and transistors.

A reverse tool helps set right the direction of current for power supplies or electrolytic capacitors by simply reversing its terminals. The tool also comes in handy to turn the circuit switch on and off, which otherwise has to be done using buttons

from the main menu. Almost all actions have an equivalent keyboard shortcut that can be exploited for faster operation.

Complex is simple. Complex circuits often require certain common design elements that are repeatedly used. Such elements can be created and exported to keep these saved. The next time these need to be used, menus help import the mini circuit and begin with the bigger design right away.

Editing the properties of components in the design easily enables a change in parameters. Any change is automatically reflected in the display window; a displayed graph is automatically recomputed for the new value and redrawn, saving a lot of trouble.

Making it easier to adapt to change. Solve Elec offers a comprehensive set of components to choose from. Properties of each of these can be configured to meet various functionalities, while staying within circuit limitations. Any component

that exceeds its maximum rating simply blows up on the screen during simulation. As real as it gets, is it not?

Handle with caution. Set the threshold value for the diode to zero for operating under ideal conditions. The same concept is extended to light emitting diodes, with the threshold now responsible for the colour of the light emitted.

Whenever an op-amp appears in the circuit, Solve Elec uses its ideal model to perform circuit analysis. Care needs to be taken with respect to the operating region of the amplifier, as once saturated, output voltage ceases to be sinusoidal and analysis is abandoned. Transistors being non-linear components are not allowed in AC designs.

Reporting the easy way

The hardest part of any design activity is the documentation that follows. A report menu in Solve Elec panel provides an easy method of selecting and combining various elements displayed in the main window, to create a ready-made report. This could include details like circuit drawings, graphs, formulae and circuit properties. There is also a provision to add editable text fields and images from other applications, and you can even have a report plan, which is similar to the contents page in manually-created ones.

Version 2.5 of Solve Elec is free to use and supports both MAC and Windows. Easy to get acquainted with and comfortable to work on, this is a useful software for beginners or advanced users in the domain of electrical engineering. The software is available on www.physicsbox.com/indexsolveelec2en.html. The same has been bundled with this month's EFY Plus DVD. Go ahead and give it a shot. **EFY**

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